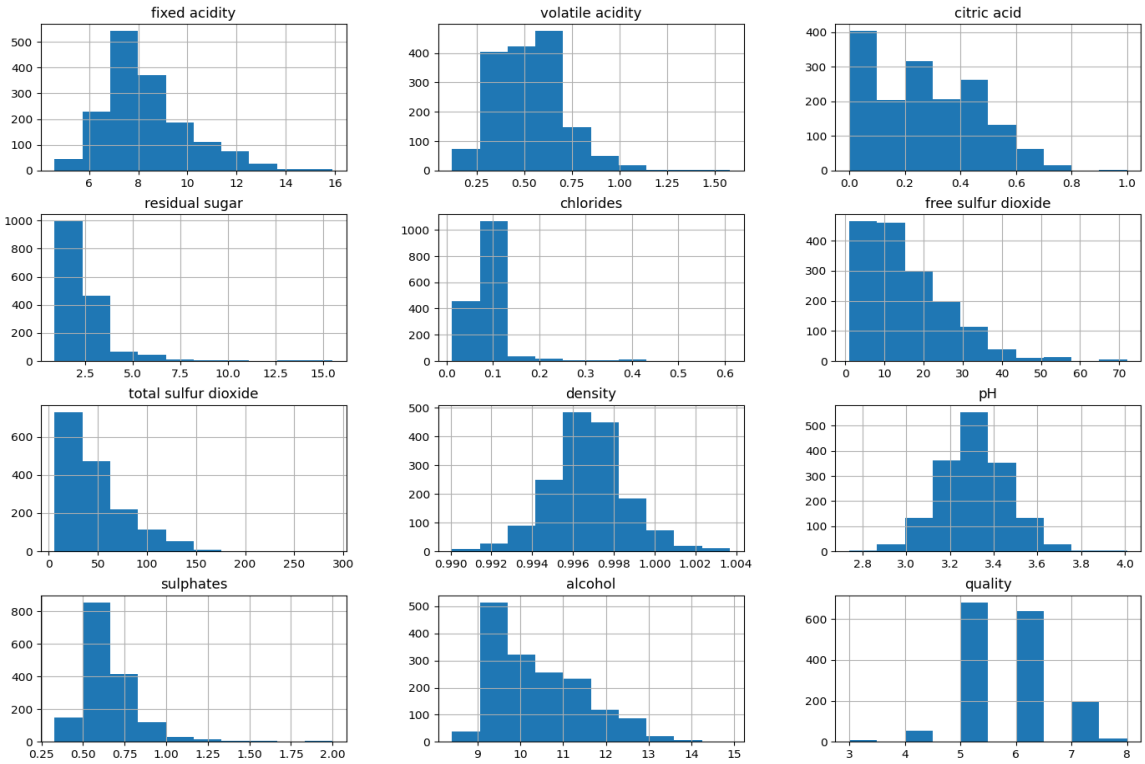
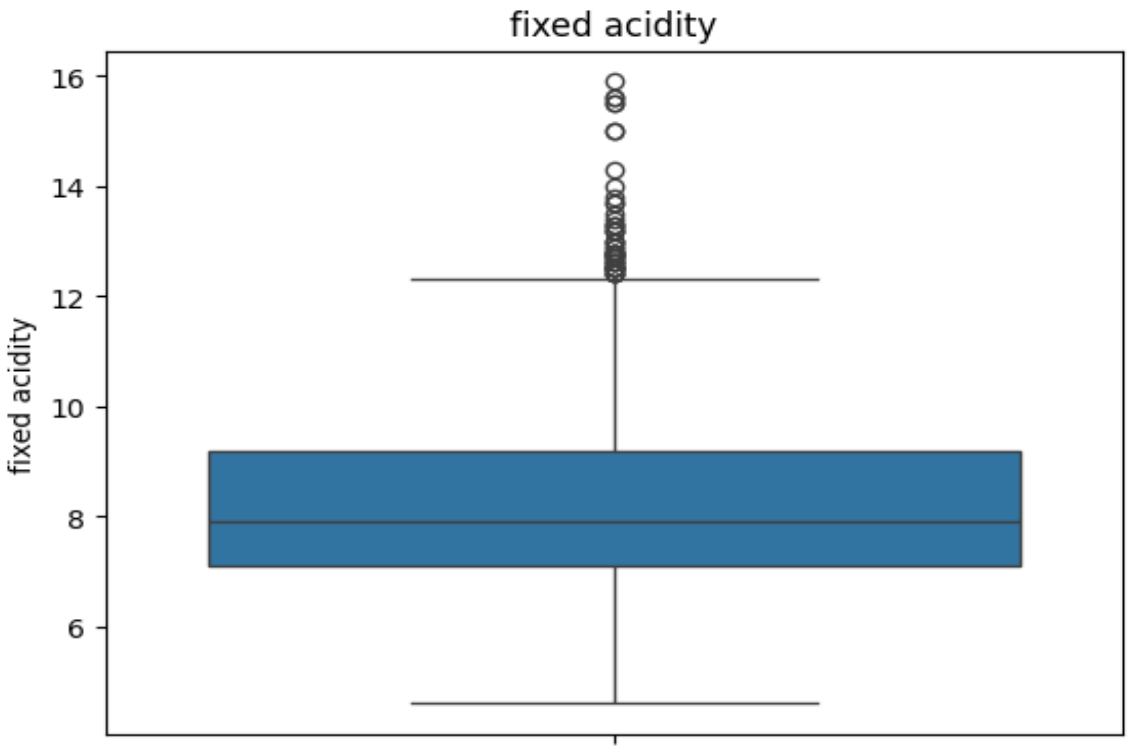
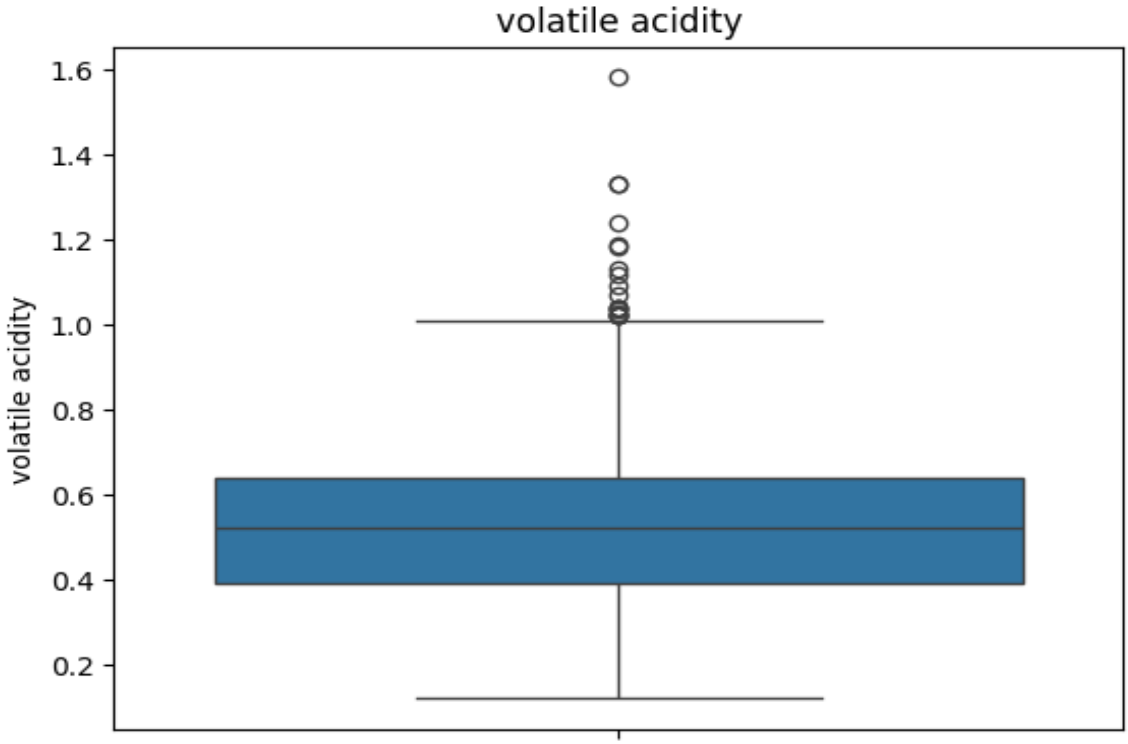
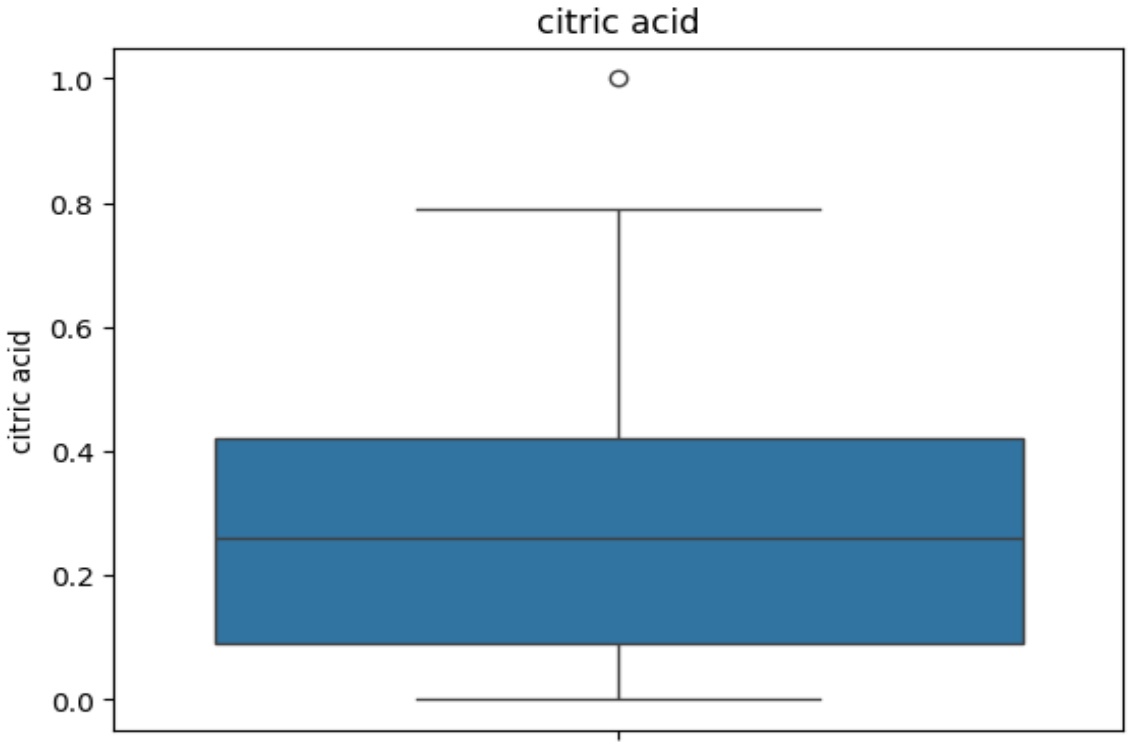
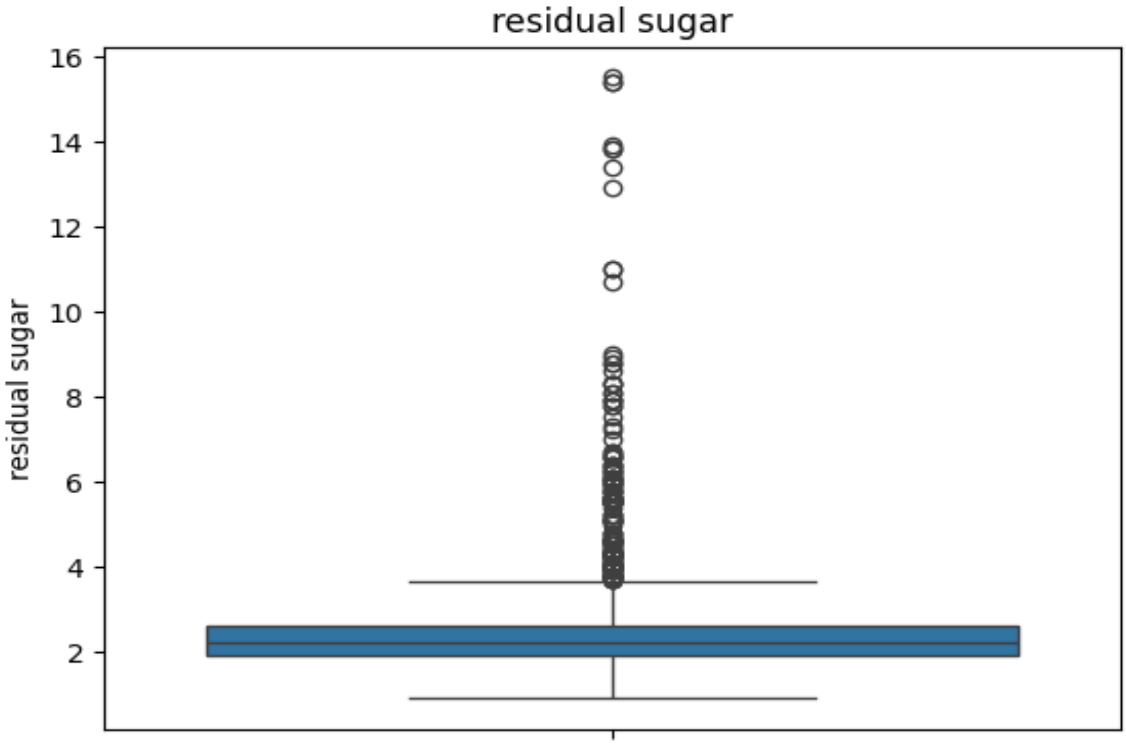
Wine Quality Project - Graphs

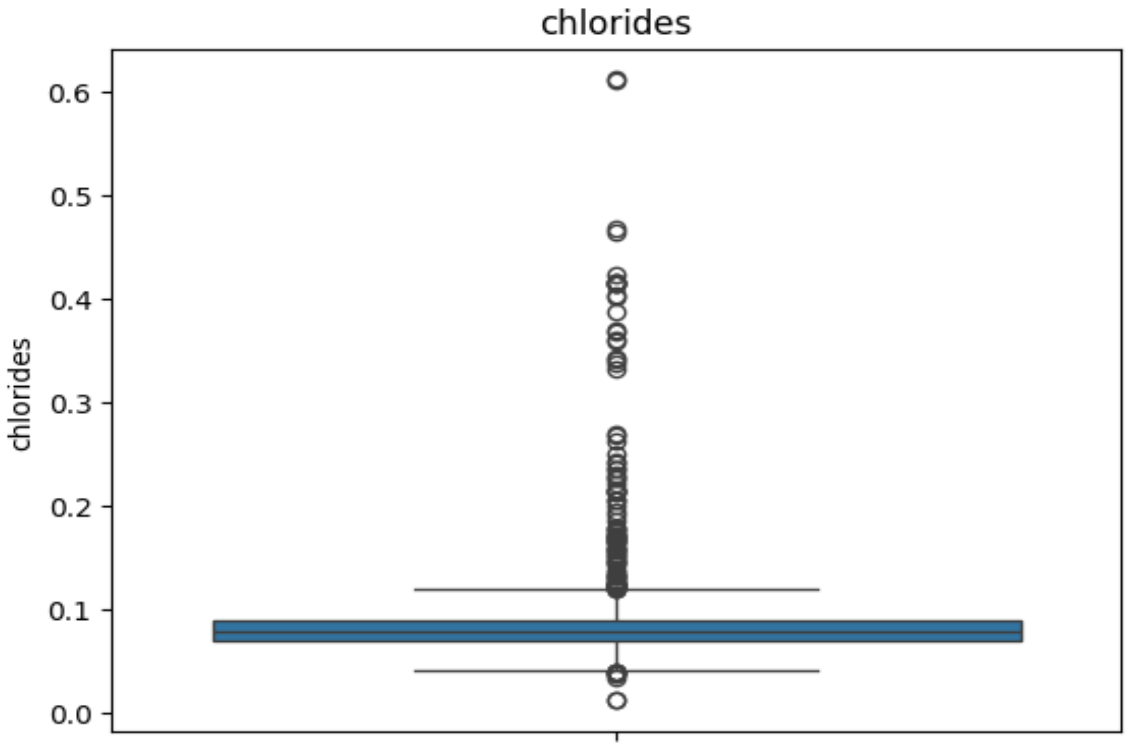


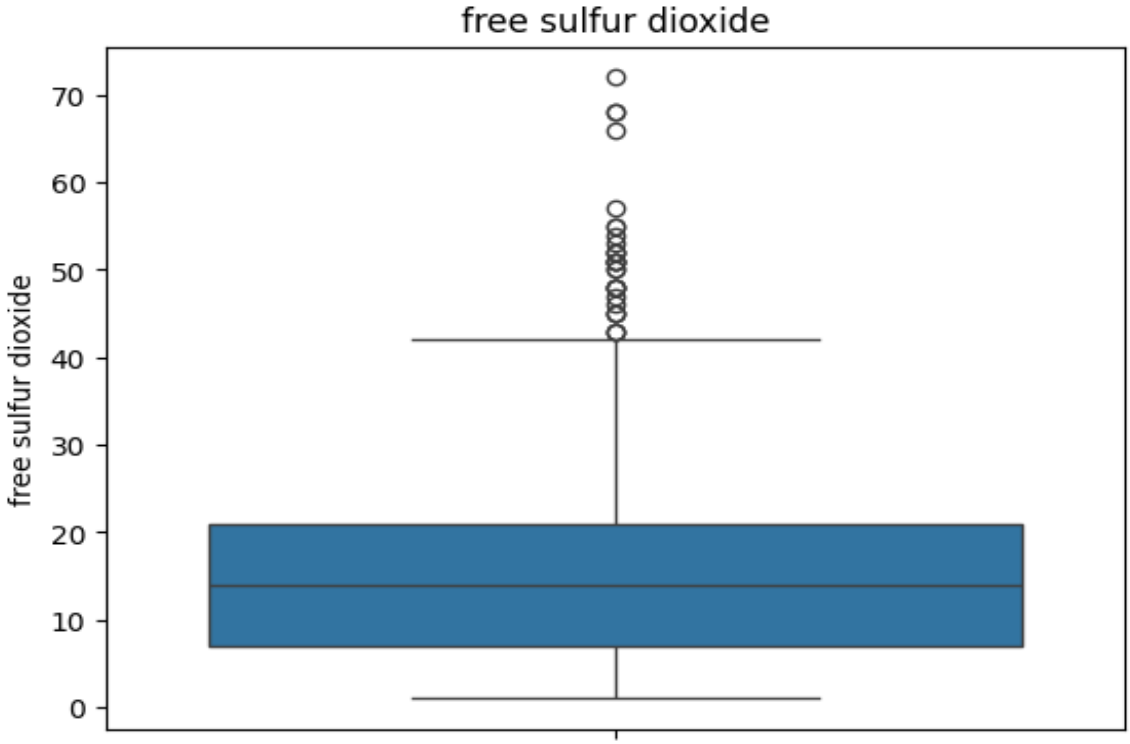


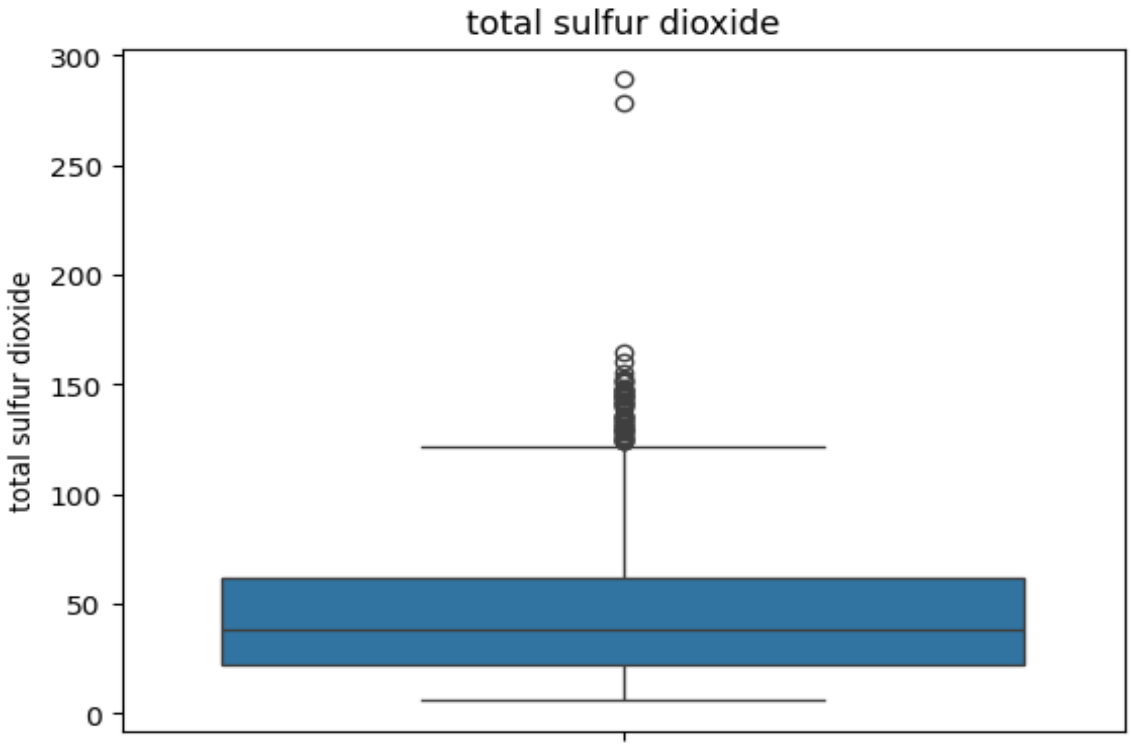

volatile acidity
volatile acidity
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2

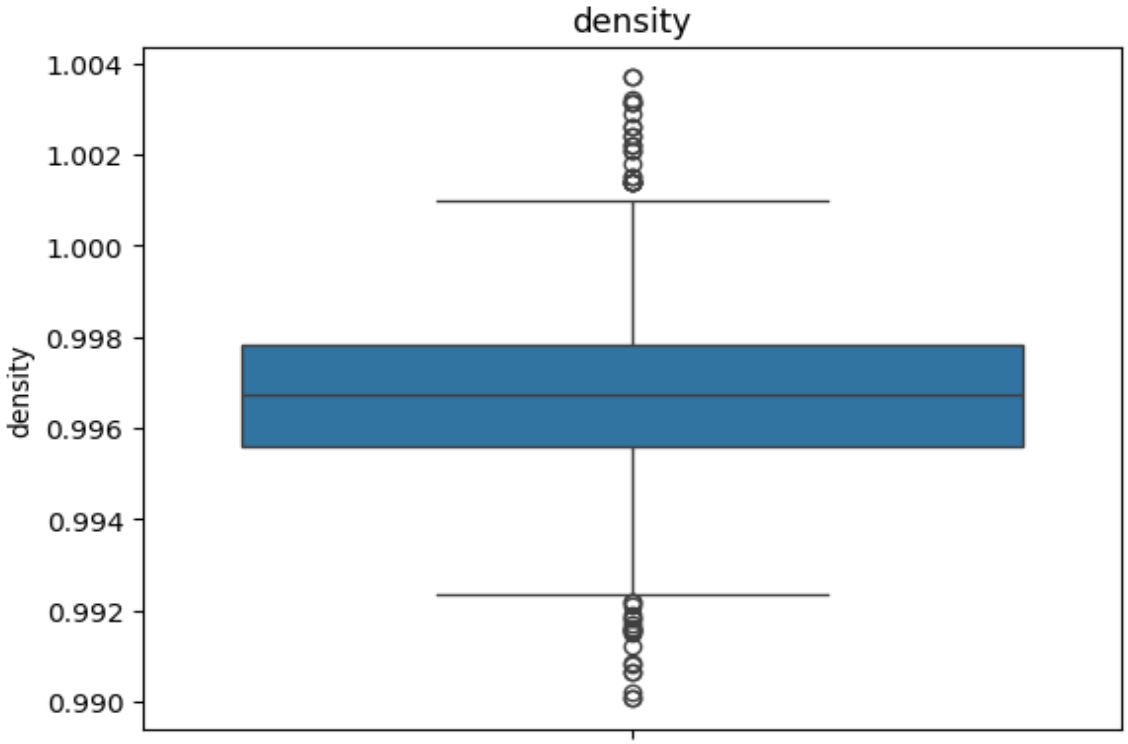

citric acid
citric acid
1.0
0.8
0.6
0.4
0.2
0.0

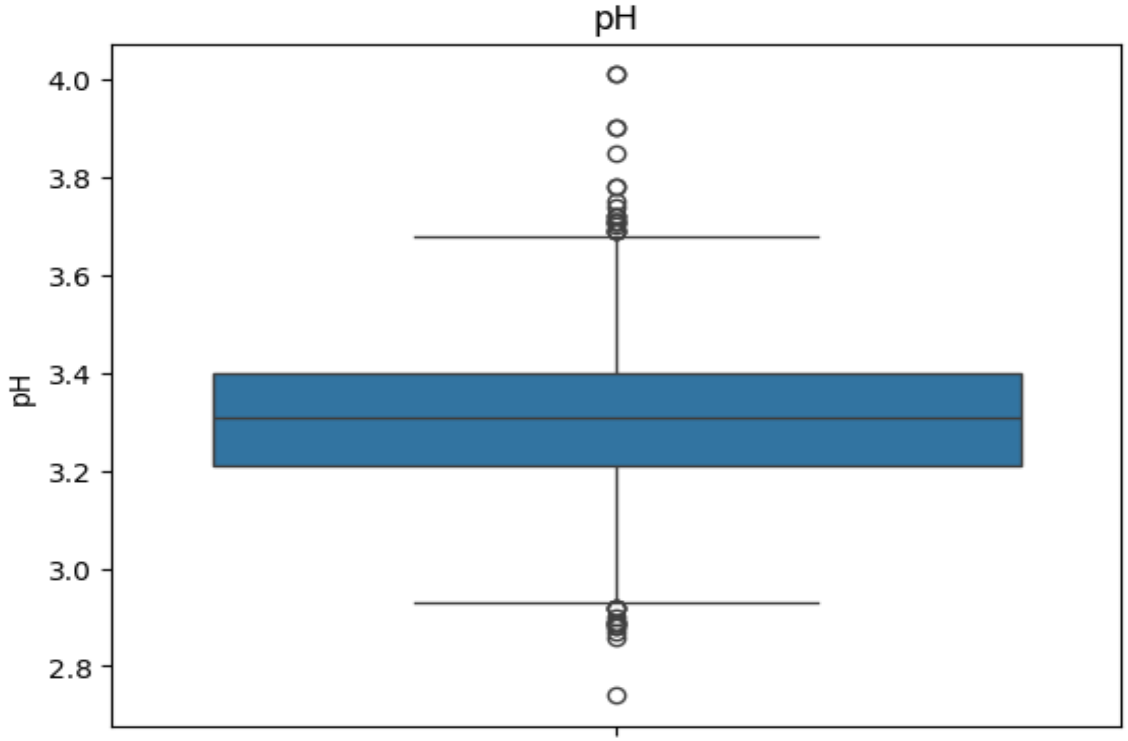

residual sugar
residual sugar
16
14
12
10
8
6
4
2

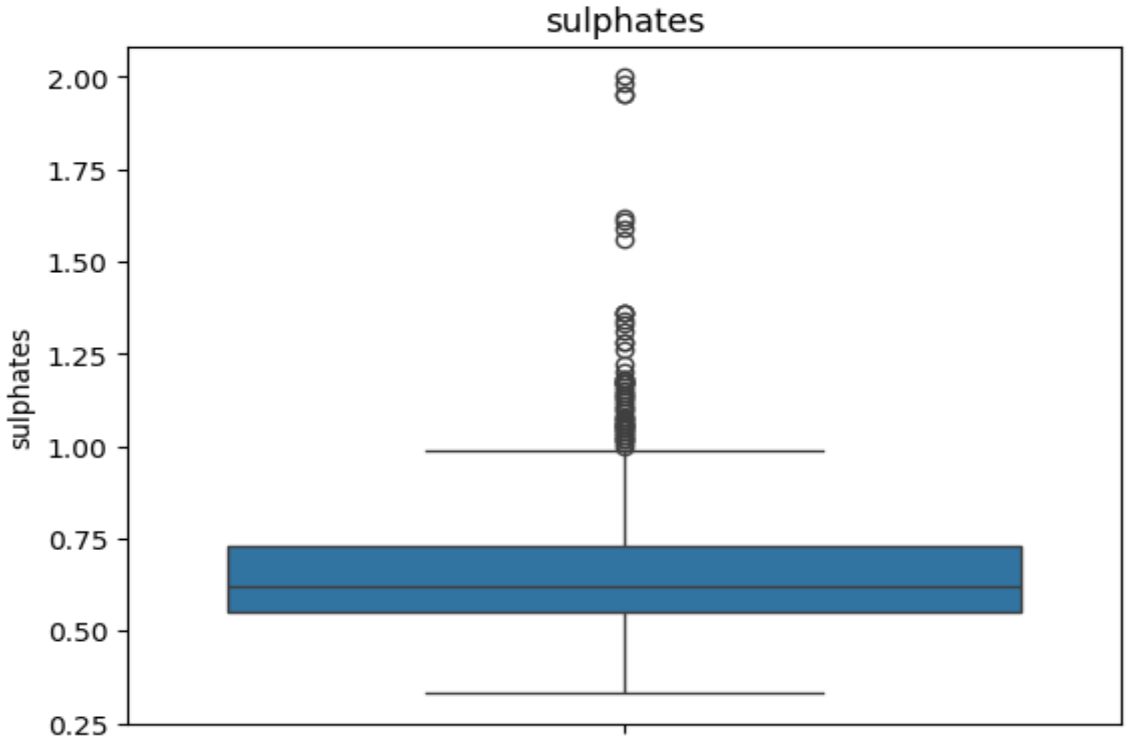

chlorides
chlorides
0.6
0.5
0.4
0.3
0.2
0.1
0.0

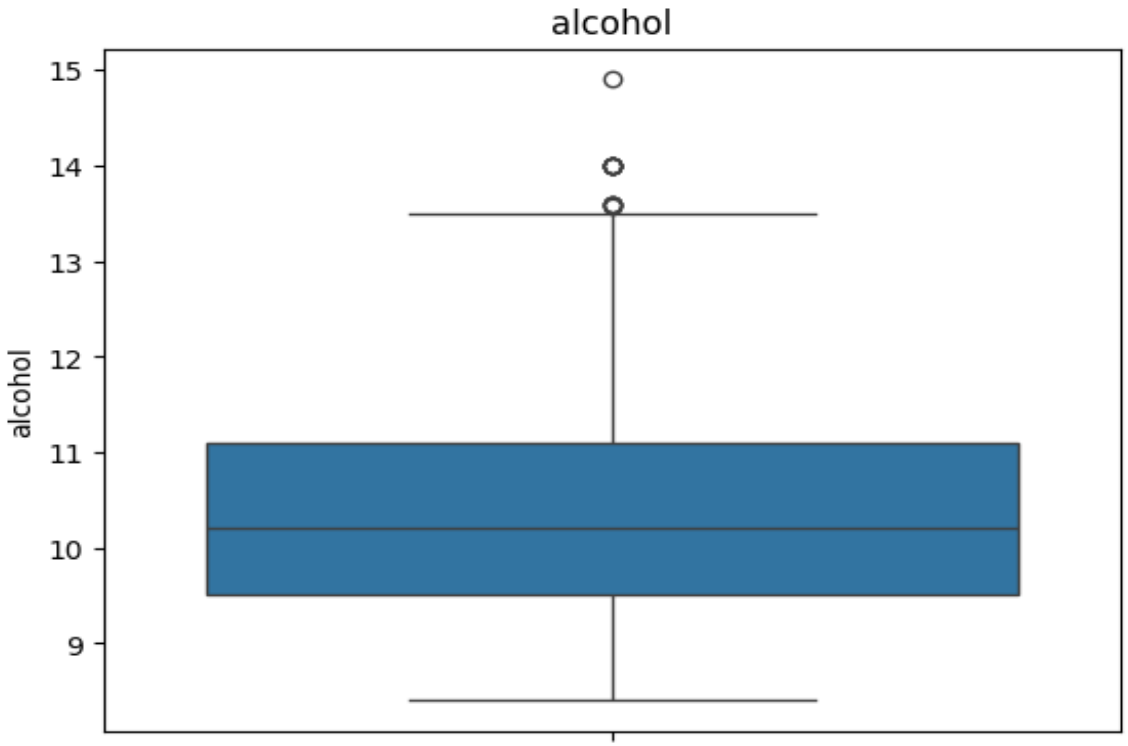

free sulfur dioxide
free sulfur dioxide
70
60
50
40
30
20
10
0

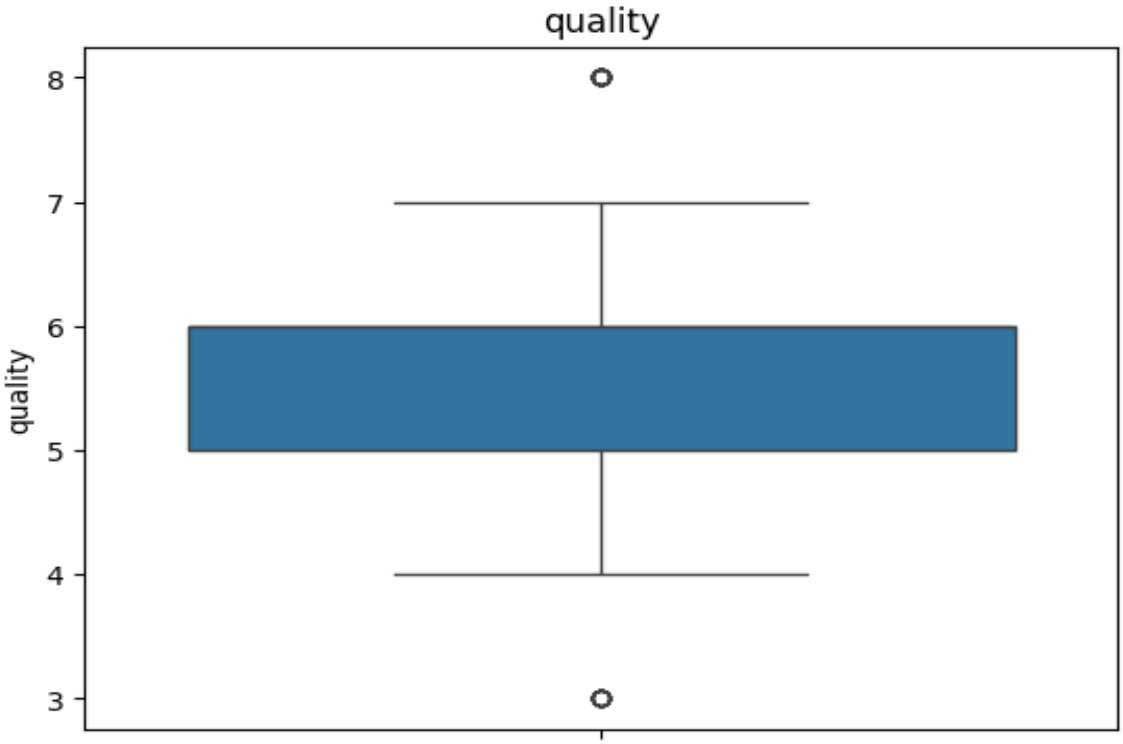

total sulfur dioxide
total sulfur dioxide
300
250
200
150
100
50
0

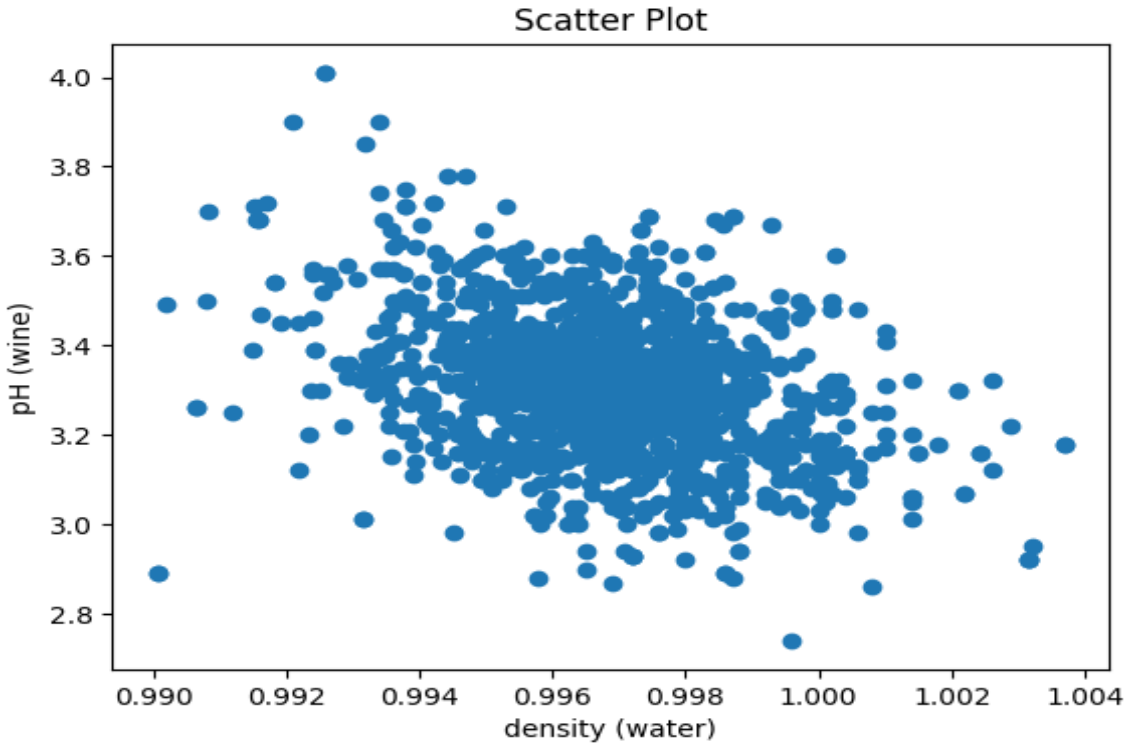

density
1.004
1.002
1.000
0.998
0.996
0.994
0.992
0.990
density

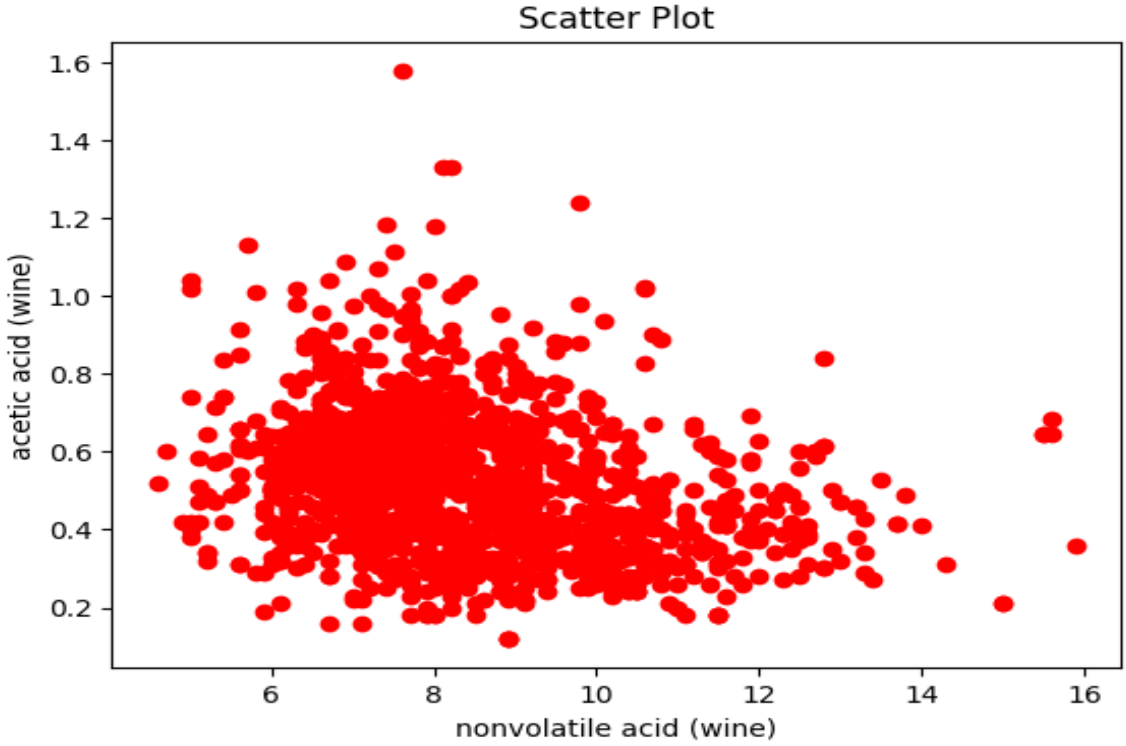

pH
4.0
3.8
3.6
3.4
3.2
3.0
2.8
pH

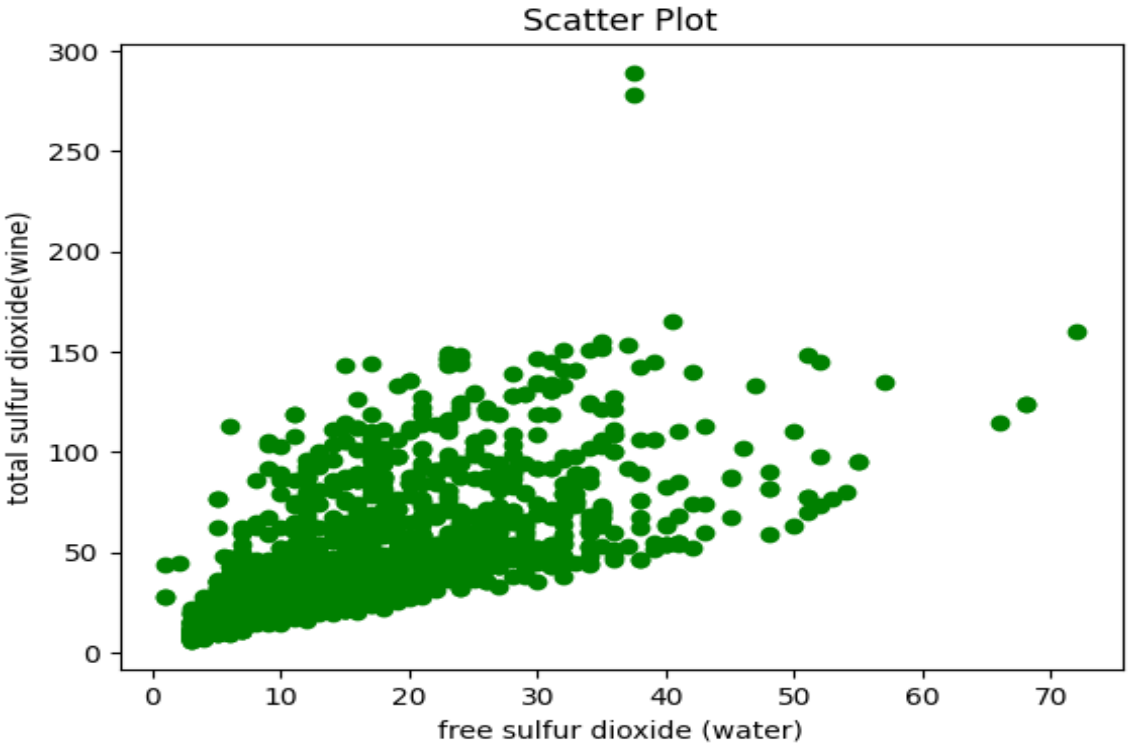

sulphates
sulphates
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25

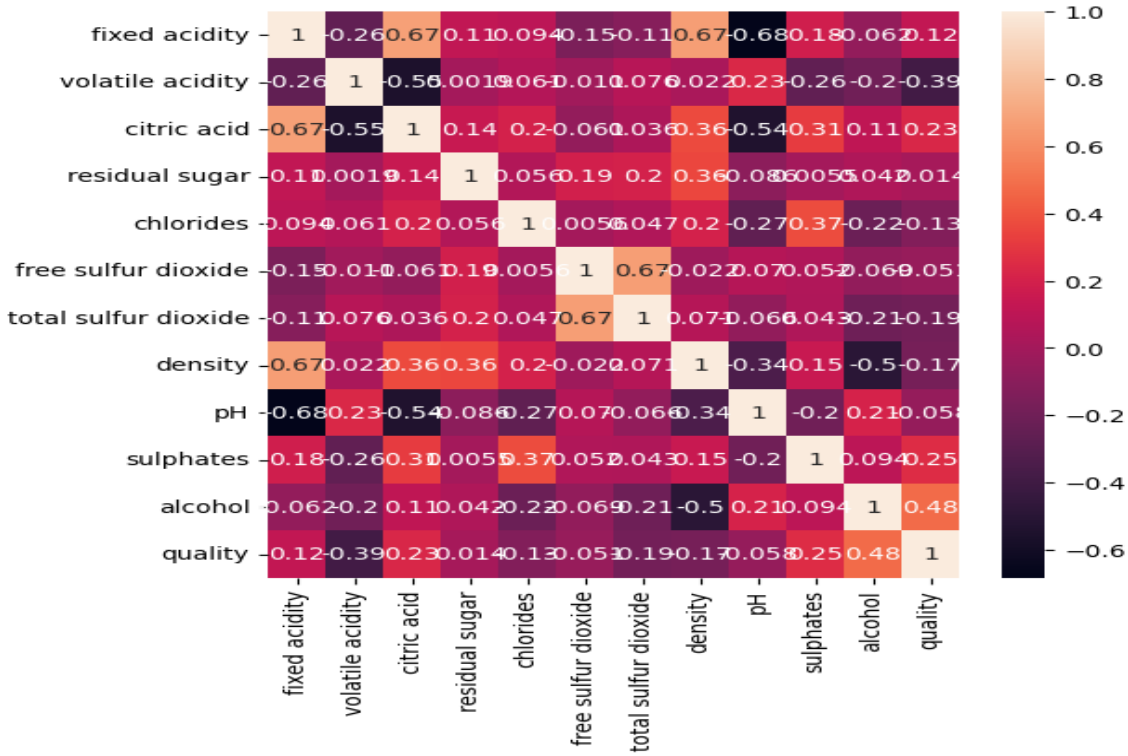

alcohol
alcohol
15
14
13
12
11
10
9

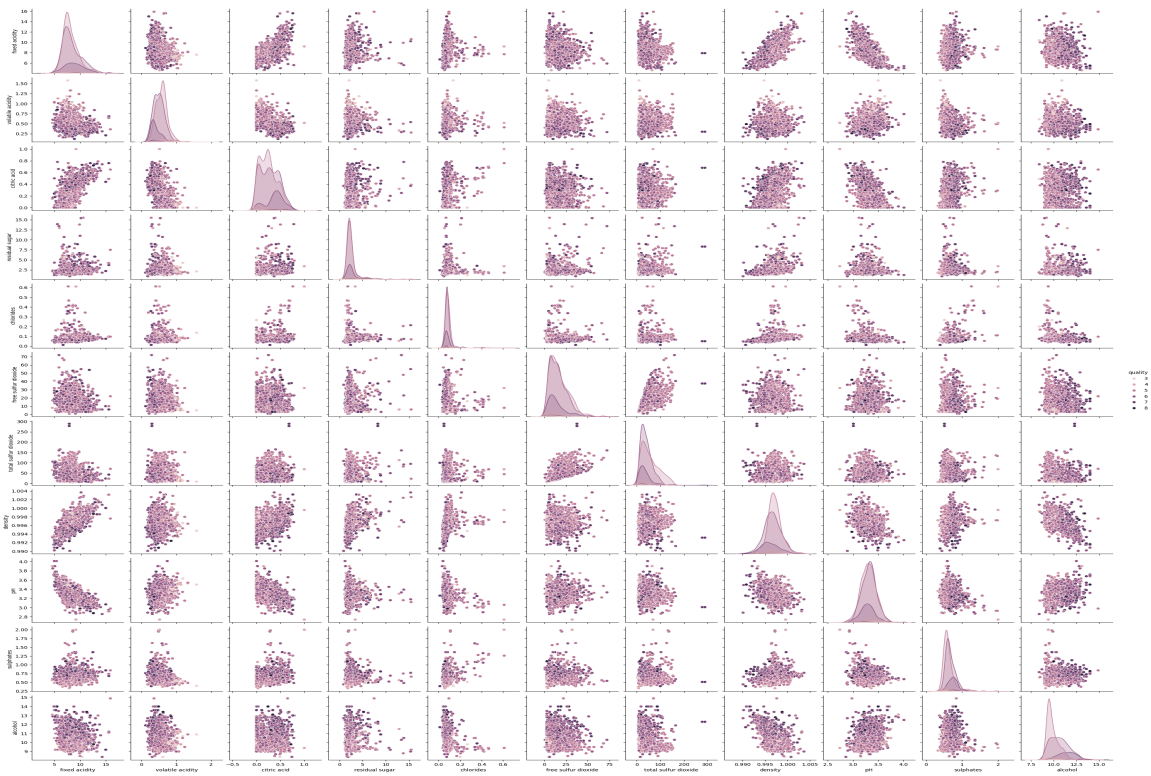

quality
8
7
6
5
4
3
quality

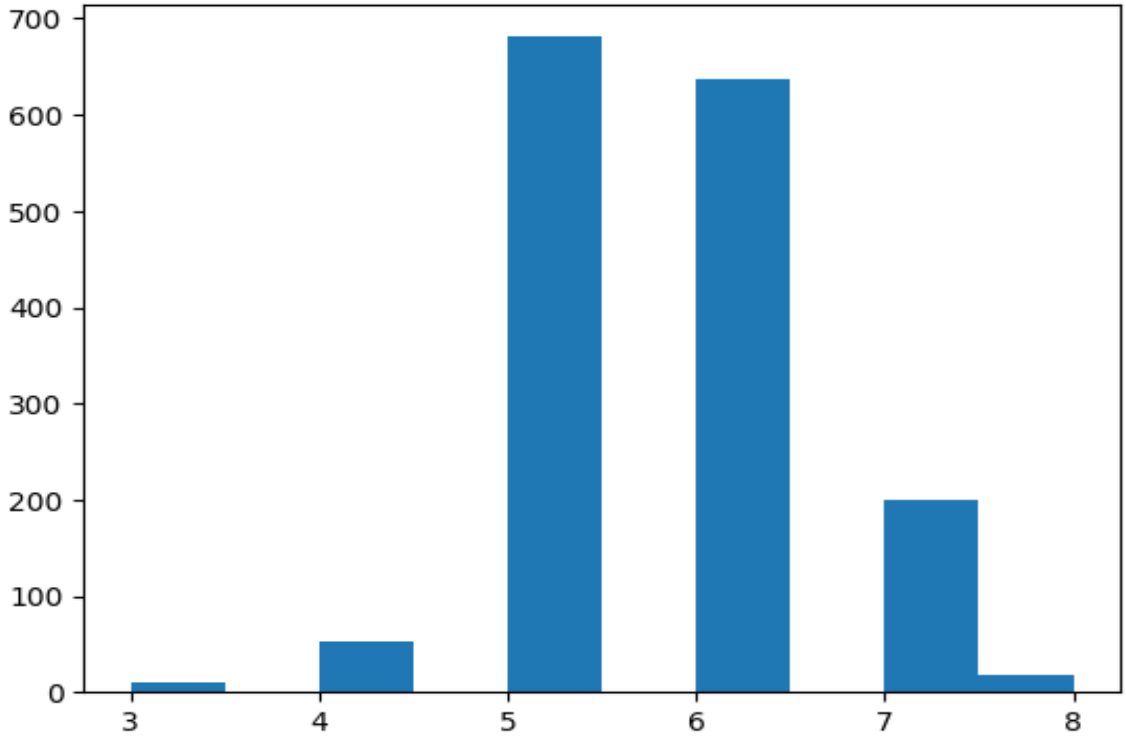

Scatter Plot
4.0
3.8
3.6
3.4
3.2
3.0
2.8
pH (wine)
0.990
0.992
0.994
0.996
0.998
1.000
1.002
1.004
density (water)

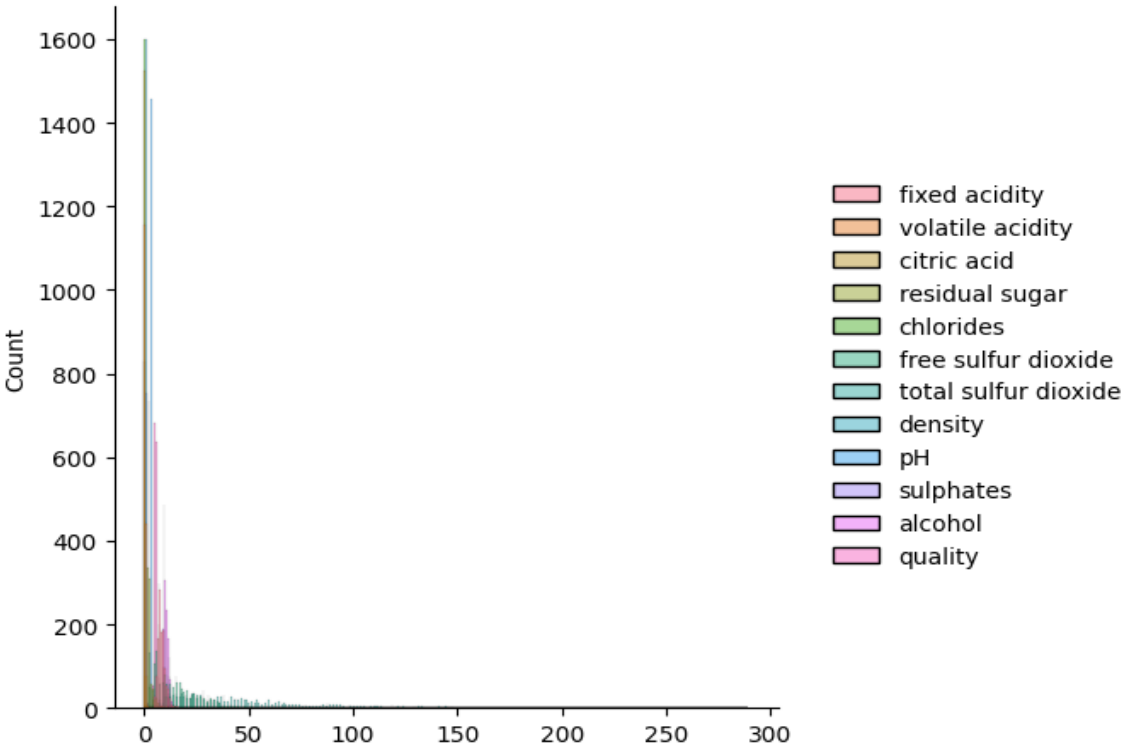

Scatter Plot
acetic acid (wine)
nonvolatile acid (wine)
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
6
8
10
12
14
16

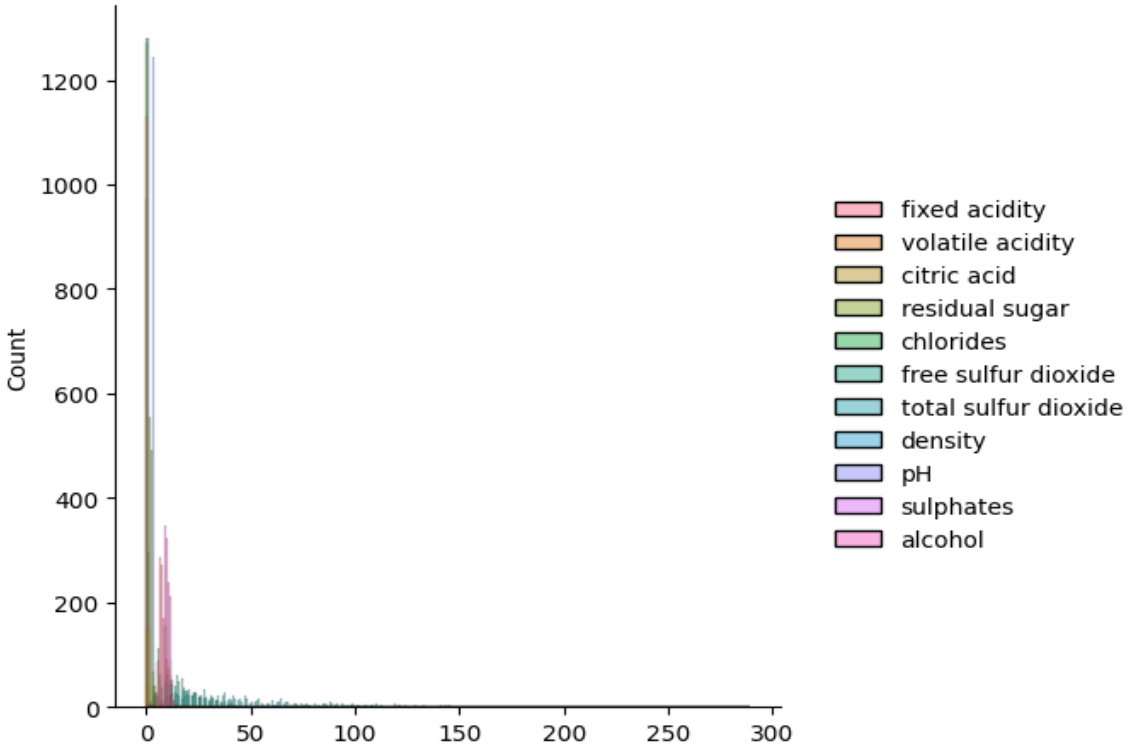

Scatter Plot
total sulfur dioxide(wine)
free sulfur dioxide (water)
300
250
200
150
100
50
0
0
10
20
30
40
50
60
70

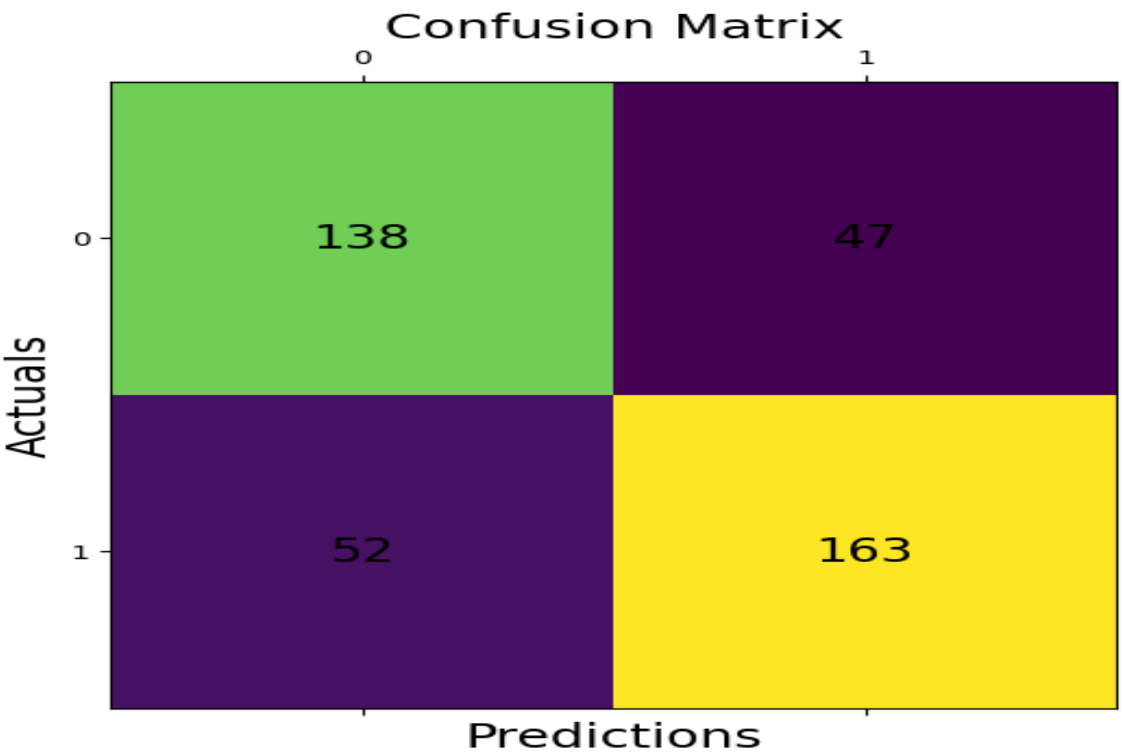

fixed acidity
volatile acidity
citric acid
residual sugar
chlorides
free sulfur dioxide
total sulfur dioxide
density
pH
sulphates
alcohol
quality
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6




700
600
500
400
300
200
100
0
3
4
5
6
7
8


1600
1400
1200
1000
800
600
400
200
0
Count
0
50
100
150
200
250
300
fixed acidity
volatile acidity
citric acid
residual sugar
chlorides
free sulfur dioxide
total sulfur dioxide
density
pH
sulphates
alcohol
quality


Count
1200
1000
800
600
400
200
0
0
50
100
150
200
250
300
fixed acidity
volatile acidity
citric acid
residual sugar
chlorides
free sulfur dioxide
total sulfur dioxide
density
pH
sulphates
alcohol


Confusion Matrix
0
1
0
1
138
47
52
163
Actuals
Predictions